

CNC Preventive Maintenance Manual

Synthetic project reference document for Phase 5 RAG testing.

1. Maintenance Philosophy

Preventive maintenance reduces unexpected downtime by inspecting components before failure. Maintenance frequency should consider manufacturer recommendations, operating hours, duty cycle, environmental conditions, and historical failure patterns.

2. Daily Inspection

Inspect machine cleanliness, coolant level and condition, lubrication level, visible leaks, tool condition, guards, cables, and abnormal sounds. Review temperature, vibration, RPM, pressure, and motor-current trends when monitoring data is available.

3. Weekly Inspection

Inspect spindle and drive components, check lubrication points, inspect belts or couplings, examine hydraulic or pneumatic connections, and verify cooling systems. Compare current observations with previous maintenance records.

4. Monthly Inspection

Perform a detailed inspection of spindle bearings, drive transmission, electrical connections, coolant system, lubrication system, and machine alignment. Check for looseness, wear, contamination, overheating, and leakage.

5. Maintenance Record Requirements

Each maintenance event should record maintenance type, description, findings, action taken, parts replaced, technician, cost, downtime, performed date, and next due date. Consistent records improve future diagnosis.

Important: This is synthetic reference content created for software testing and demonstration. It is not a manufacturer manual or a substitute for approved maintenance procedures, qualified personnel, or applicable safety requirements.